

Department Overview Department Name: Department of Software Engineering Faculty: Faculty of Information & Communication Technology (FICT) Accreditation: The program is accredited by the Pakistan Engineering Council (PEC). Core Focus: Combines theoretical and practical software engineering principles. It encompasses the systematic, disciplined, and quantifiable approach to the design, development, operation, and maintenance of large-scale, reliable, efficient, and economical software systems. • History: The department has been offering a four-year undergraduate software engineering degree since 2012 to meet growing national and international market demands. • Specialization Fields: Requirements engineering, quality engineering, software testing, and project management.

Vision and Mission Vision: To attain excellence in education and research at the national and international level and produce technically, professionally, and ethically sound graduates.

Mission: To produce the next-generation Software Engineering graduates who are fully equipped with concrete software engineering knowledge, real-world problem-solving skills, research abilities, excellent leadership qualities, and entrepreneurial abilities required for their future careers, and serving their communities.

Bachelor of Science in Software Engineering (BSSE)

Admission Requirements Academic Background: F.Sc (Pre-Engineering) / ICS with Mathematics and Physics from any recognized board (or equivalent qualification) with at least 60% marks. Alternatively, a Diploma of Associate Engineer (DAE) in a relevant field with a minimum of 60% marks.

Degree Requirements Total Credit Hours: 132 Course Structure: 42 courses plus the Final Year Design Project (FYDP Part I & II). Minimum Academic Standing: CGPA $\geq$ 2.0

Program Educational Objectives (PEOs) The department targets training students across all aspects of the software development lifecycle to achieve the following career and professional objectives:

1. Utilize critical-thinking and engineering abilities to excel in the domain of Software Engineering.
2. Act as professional, ethical, and knowledgeable citizens.
3. Demonstrate leadership and work collaboratively in diverse teams and organizations.

Curriculum Plan (Program Schema by Semester) Note: Credit hour listings follow the format (Theory-Lab).

1st Semester • IT-102 | Information and Communication Technology (ICT) | Prerequisite: NIL | Credit Hours: 2-1 • SE-2XX | Software Engineering Fundamentals | Prerequisite: NIL | Credit Hours: 3-0 • ENG-161 | Reading and Writing Skills | Prerequisite: None | Credit Hours: 3-0 • PHY-205 | Applied Physics | Prerequisite: NIL | Credit Hours: 2-1 • HUM-138 / HUM-1XX | Psychology / Islamic Studies and Ethics | Prerequisite: NIL/NIL | Credit Hours: 2-0 / 2-0 • XX-1XX | Occupational Health and Safety | Prerequisite: NIL | Credit Hours: 1-0

2nd Semester • CS-114 | Programming Fundamentals | Prerequisite: NIL | Credit Hours: 3-1 • ENG-361 | Communication & Presentation Skills | Prerequisite: NIL | Credit Hours: 3-0 • MATHA-234 | Discrete Structures | Prerequisite: NIL | Credit Hours: 3-0 • MATHP-118 | Calculus & Analytical Geometry | Prerequisite: NIL | Credit Hours: 3-0 • CS-119 / HUM-102 | Web Development (WD) / Pakistan Studies & Global Perspective | Prerequisite: NIL/NIL | Credit Hours: 2-1 / 2-0

3rd Semester • CS-212 | Object-Oriented Programming | Prerequisite: Programming Fundamentals | Credit Hours: 3-1 • STAT-101 | Probability & Statistics | Prerequisite: NIL | Credit Hours: 3-0 • CE-2XX | Computer Architecture and Logic Design | Prerequisite: NIL | Credit Hours: 3-1 • MATHP-111 | Linear Algebra | Prerequisite: NIL | Credit Hours: 3-0 • CE-302 | SE Elective I (Web Engineering (WE)) | Prerequisite: Web Development | Credit Hours: 2-1

4th Semester • CS-214 | Data Structures and Algorithms | Prerequisite: Object-Oriented Programming | Credit Hours: 3-1 • SE-231 | Software Design & Architecture | Prerequisite: Software Engineering Fundamentals | Credit Hours: 2-1 • IT-3XX | Human-Computer Interaction | Prerequisite: NIL | Credit Hours: 2-1 • XX-XXX | SE Elective - II | Prerequisite: NIL | Credit Hours: 3-0 • MGMT-206 / MATHA-213 | Principles of Management / Complex Variables & Transform | Prerequisite: NIL/NIL | Credit Hours: 3-0 / 2-0

5th Semester • SE-332 | Software Construction and Development | Prerequisite: Software Design & Architecture | Credit Hours: 2-1 • MGMT-2XX | Entrepreneurship and Marketing | Prerequisite: NIL | Credit Hours: 2-0 • XX-XXX | SE Elective - III | Prerequisite: NIL | Credit Hours: 3-0 • CS-332 / CS-341 | Database Systems / Operating Systems | Prerequisite: NIL/NIL | Credit Hours: 3-1 / 3-1

6th Semester • SE-321 | Software Quality Engineering | Prerequisite: Software Engineering Fundamentals | Credit Hours: 3-0 • TE-307 | Computer Networks | Prerequisite: NIL | Credit Hours: 3-1 • MATHA-216 | Numerical Analysis | Prerequisite: NIL | Credit Hours: 2-1 • XX-XXX | MDEE-I | Prerequisite: NIL | Credit Hours: 3-0 • HUM-266 | Technical & Business Writing | Prerequisite: NIL | Credit Hours: 3-0

7th Semester • SE-352 | Software Project Management | Prerequisite: NIL | Credit Hours: 3-0 • IT-321 | Information Security | Prerequisite: NIL | Credit Hours: 3-0 • SE-491L | Senior Design Project - I (FYDP-I) | Prerequisite: NIL | Credit Hours: 0-3 • XX-XXX | SE Elective - IV | Prerequisite: NIL | Credit Hours: 3-0 • CS-4XX | Design and Analysis of Algorithms | Prerequisite: Data Structures and Algorithms | Credit Hours: 3-0

8th Semester • XX-XXX | MDEE-II | Prerequisite: NIL | Credit Hours: 3-0 • SE-492L | Senior Design Project - II (FYDP-II) | Prerequisite: Senior Design Project - I | Credit Hours: 0-3 • XX-XXX / XX-XXX | SE Elective - V / SE Elective VI | Prerequisite: NIL/NIL | Credit Hours: 3-0 / 3-0 • SE-471 | Formal Methods in Software Engineering | Prerequisite: NIL | Credit Hours: 3-0